



**ECE 639**  
**Selected Topics in Electronics**  
**Power Management**  
**Spring 2021**

**Topic 2**  
**Voltage & Current References**

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**(Some slides are courtesy of Edgar Sanchez-Sinencio, Texas A&M  
and Tao Li et al., Penn State)**

# Outline

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- **The Need for BGR**
- **BGR Basic Concept**
- **BGR Topologies**
  - Brokaw Cell
  - BGR with Opamp
  - Sub-bandgap BGR
  - Low-power BGR
- **Higher-order Curvature Compensation**
- **Startup Circuit**
- **BGR Trimming**
- **BGR Specifications**
- **Current References**



# The Need for BGR (1)

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**Negative-feedback system requires stable reference**



# The Need for BGR (2)

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- **An IC should work**
  - Even when  $V_{DD}$  changes (Supply variation) **V**
    - $V_{DD} \pm 10\%$
  - Even when temperature changes (Temperature variation) **T**
    - $-40\text{ }^{\circ}\text{C} - 125\text{ }^{\circ}\text{C}$
  - Even when physical properties change (Process variation) **P**
    - BJTs:  $\beta$ :  $\pm 30\%$
    - MOS:  $\mu$ :  $\pm 10\%$ ,  $V_{th}$ :  $\pm 100\text{mV}$
    - Resistors:  $R$ :  $\pm 20\%$
    - Capacitors:  $C$ :  $\pm 15\%$
    - Inductors:  $L$ :  $\pm 2\%$



# BGR Basic Concept (1)

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- Add two quantities with opposite temperature coefficient with proper weighting → The resultant quantity exhibits zero temperature coefficient.

$$V_{REF} = C_1 V_1 + C_2 V_2$$

$$\frac{\partial V_{REF}}{\partial T} = 0$$

$$C_1 \frac{\partial V_1}{\partial T} + C_2 \frac{\partial V_2}{\partial T} = 0$$

$$\therefore \frac{\partial V_1}{\partial T} = -\frac{C_2}{C_1} \frac{\partial V_2}{\partial T}$$

**+ve TC & -ve TC**



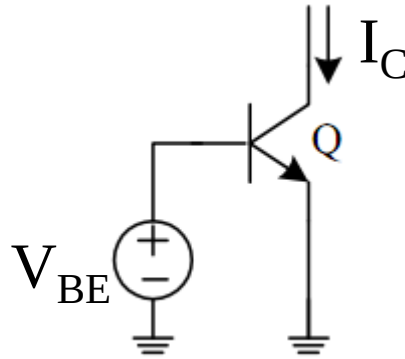
# BGR Basic Concept (2)

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- The characteristics of PN junctions (BJTs and Diodes) have proven the most well defined quantities providing positive and negative TC.
- $kT/q$  has a positive temperature coefficient.
  - "PTAT" proportional to absolute temperature
- $V_{BE}$  of a BJT or  $V_D$  of a diode decrease with temperature
  - "CTAT" complementary to absolute temperature
- Can combine PTAT + CTAT to yield an approximately zero TC voltage reference



# Thermal Behavior of a BJT or a Diode



$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

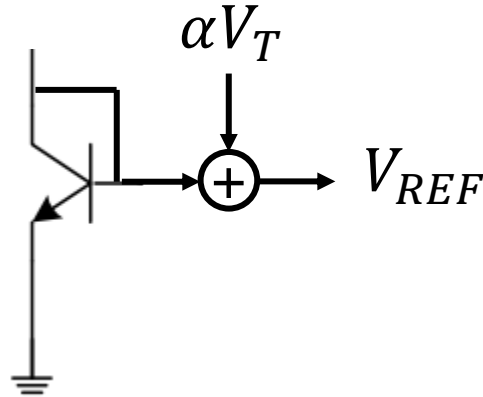
$$V_{BE} = V_T \ln \left( \frac{I_C}{I_S} \right) = \frac{kT}{q} \ln \left( \frac{I_C}{I_S} \right)$$

$$\text{But } I_S = bT^{2.5} e^{\frac{-E_g}{kT}}$$

$$\text{This gives } \frac{\partial V_{BE}}{\partial T} \approx -2 \text{ mV/K @ 300 K (CTAT)}$$



# The Need for a PTAT



$$V_{REF} = V_{BE} + \alpha V_T$$

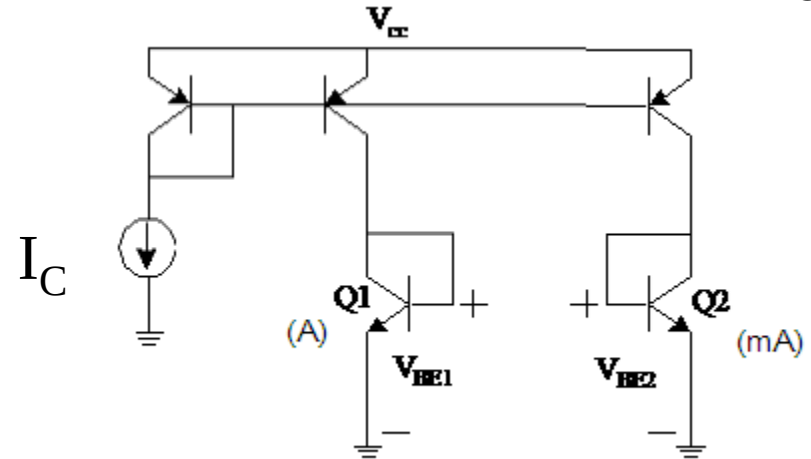
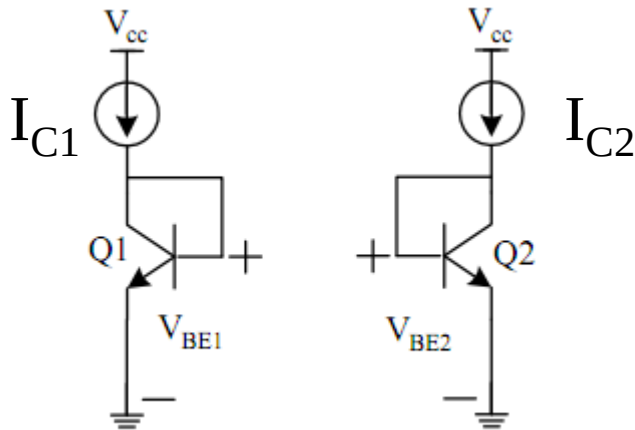
$$TC = -2 \frac{\text{mV}}{\text{K}} + 0.085\alpha$$

$$TC = 0 \text{ when } \alpha \approx 23$$

**This gives  $V_{REF} = 1.205 \text{ V}$  (Bandgap of Silicon at 0 K)**

# PTAT Generation

- Amplifying the difference in  $V_{BE}$  of two BJTs  $\rightarrow$  PTAT term
- Different  $V_{BE}$  voltages can be obtained by
  - Applying different  $I_C$
  - Using two BJT's with different emitter areas but equal  $I_C$



$$\Delta V_{BE} = V_{BE1} - V_{BE2}$$

$$\Delta V_{BE} = V_T \ln \left( \frac{I_{C1}}{I_{C2}} \right) = \frac{kT}{q} \ln \left( \frac{I_{C1}}{I_{C2}} \right)$$

$$\Delta V_{BE} = V_{BE1} - V_{BE2}$$

$$\Delta V_{BE} = V_T \ln \left( \frac{I_{S2}}{I_{S1}} \right) = V_T \ln(m)$$

# Brokaw Cell

$$V_{REF} = V_{BE1} + V_{R2} = V_{BE2} + V_{R1} + V_{R2}$$

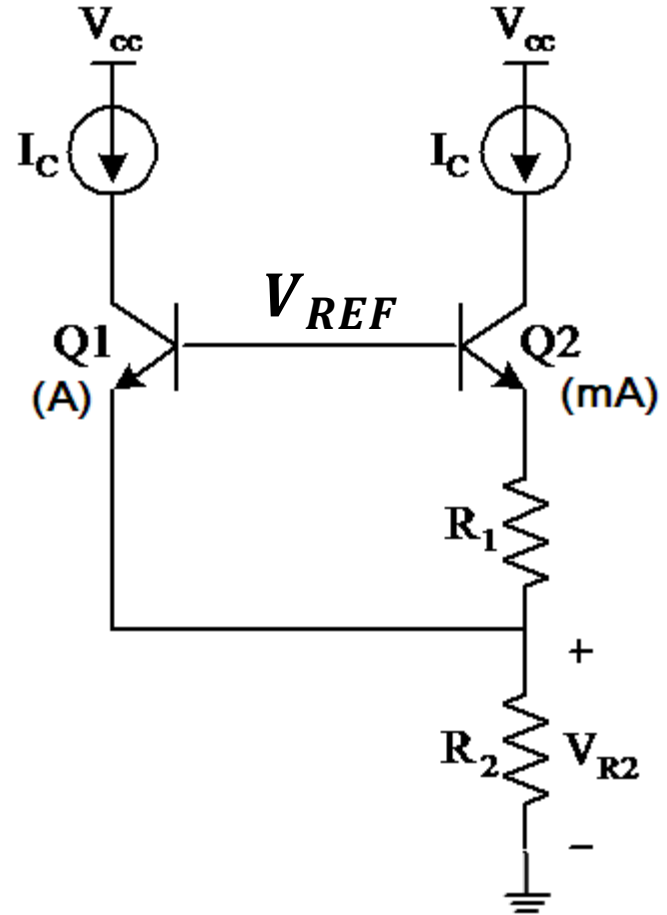
$$V_{BE1} = V_{BE2} + V_{R1}$$

$$V_{R1} = V_{BE1} - V_{BE2} = V_T \ln(m)$$

$$V_{R1} = I_C R_1$$

$$V_{R2} = 2I_C R_2 = 2V_{R1} \frac{R_2}{R_1} = 2 \frac{R_2}{R_1} V_T \ln(m)$$

$$V_{REF} = V_{BE1} + 2 \frac{R_2}{R_1} V_T \ln(m)$$



# BGR with Opamp (1)

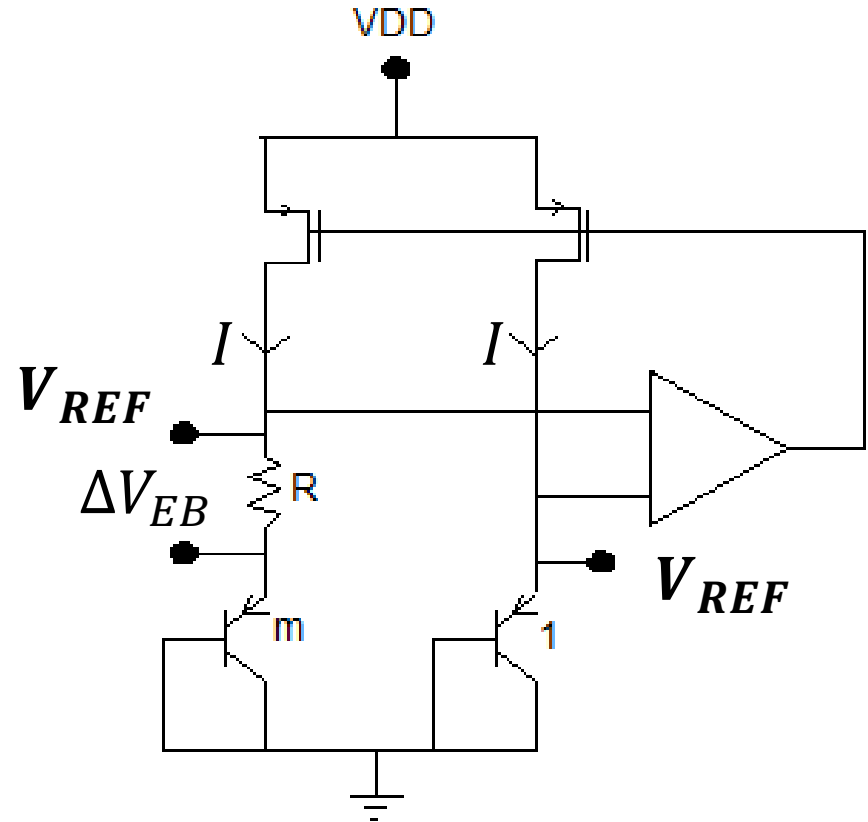
$$V_{REF} = V_{EB1} = V_{EB2} + V_R$$

$$\Delta V_{EB} = V_{EB1} - V_{EB2} = IR$$

$$\Delta V_{EB} = V_T \ln(m)$$

$$V_{REF} = V_{EB2} + V_T \ln(m)$$

- **Positive or negative FB?**



# BGR with Opamp (2)

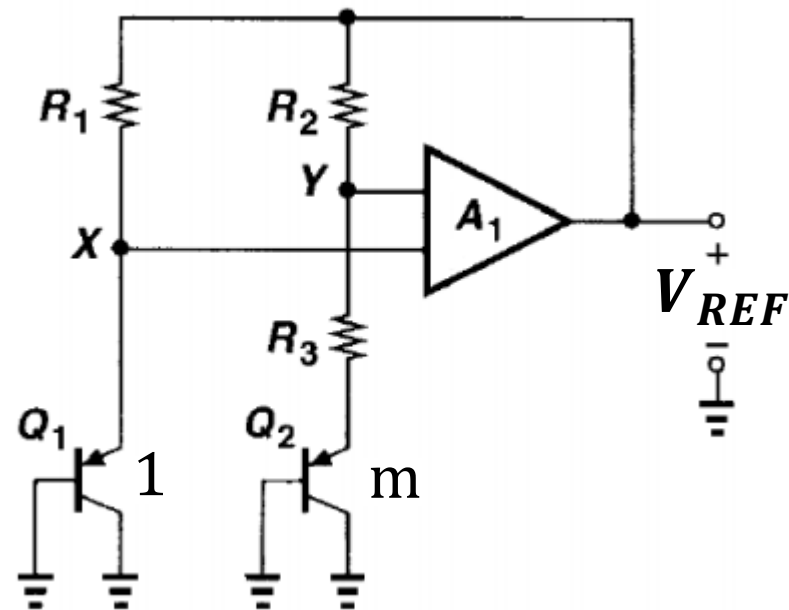
$$V_{REF} = V_{EB1} + IR_1 = V_{EB2} + I(R_2 + R_3)$$

$$V_X = V_Y$$

$$\Delta V_{EB} = V_{EB1} - V_{EB2} = IR_3$$

$$I = \frac{V_T}{R_3} \ln(m)$$

$$V_{REF} = V_{EB1} + V_T \ln(m) \frac{R_1}{R_3}$$



- Positive or negative FB?

# Sub-bandgap BGR

$$V_{REF} = I_o R_o = (I_1 + I) R_o$$

$$I_1 = \frac{V_{EB1}}{R_1}$$

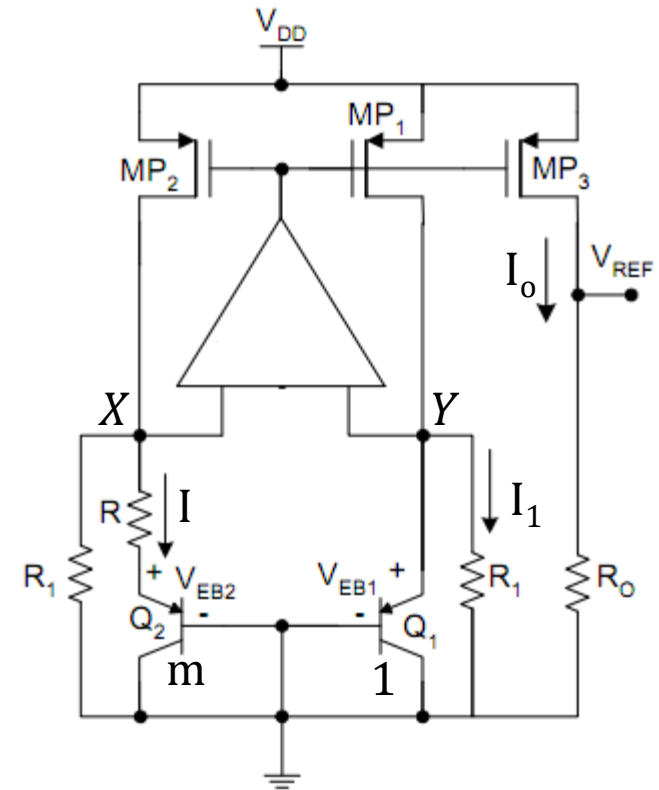
$$V_X = V_Y$$

$$\Delta V_{EB} = V_{EB1} - V_{EB2} = IR$$

$$I = \frac{V_T}{R} \ln(m)$$

$$V_{REF} = V_{EB1} \frac{R_o}{R_1} + V_T \ln(m) \frac{R_o}{R}$$

- Positive or negative FB?



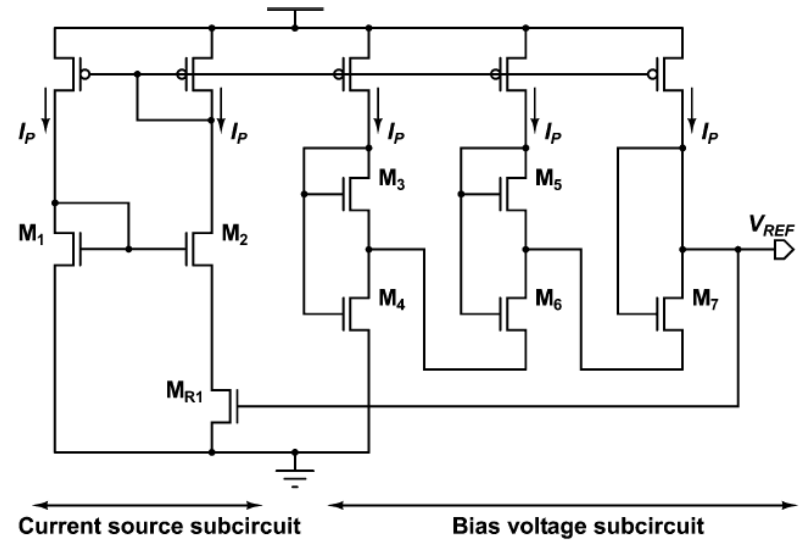
# All-MOSFET BGR

- All MOSFETs but  $M_{R1}$  are operating in Sub-threshold.
- $M_{R1}$  is operated in a strong-inversion, deep-triode region.
- $V_{GS1} = V_{GS2} + V_{R1}$
- But  $I_1 = I_2 = I_p$
- $V_{R1} = \eta V_T \ln(W_2/W_1)$

$$I_P = \frac{V_{DSR1}}{R_{M_{R1}}} = K_{R1} \mu C_{OX} (V_{REF} - V_{TH}) \eta V_T \ln(K_2/K_1)$$

$$\begin{aligned} V_{REF} &= V_{GS4} - V_{GS3} + V_{GS6} - V_{GS5} + V_{GS7} \\ &= V_{GS4} + \eta V_T \ln \left( \frac{2K_3K_5}{K_6K_7} \right) \\ &= \textcircled{V_{TH}} + \eta V_T \ln \left( \frac{3I_P}{K_4I_0} \right) + \eta V_T \ln \left( \frac{2K_3K_5}{K_6K_7} \right) \end{aligned}$$

CTAT



[Ueno JSSC09]

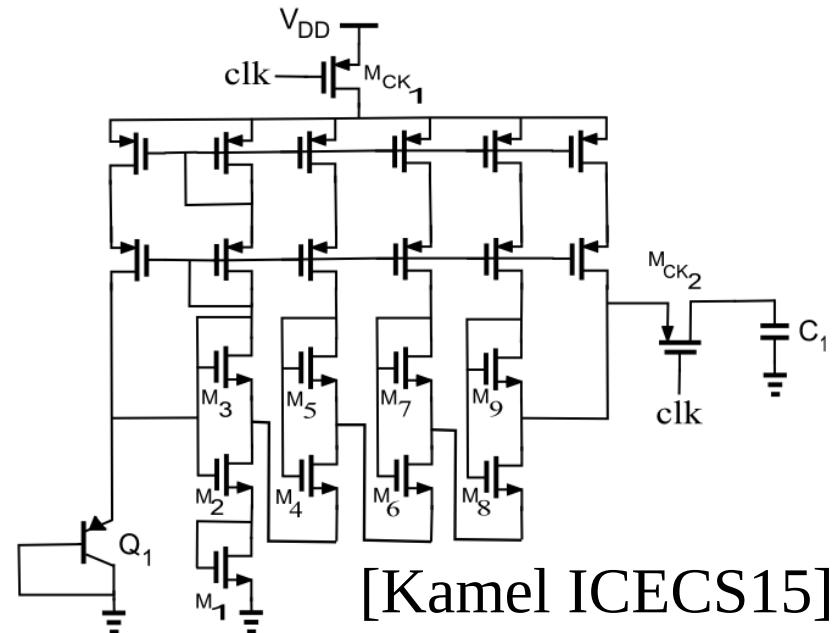
$$I_D = K I_0 \exp \left( \frac{V_{GS} - V_{TH}}{\eta V_T} \right)$$

$$I_0 = \mu C_{OX} (\eta - 1) V_T^2$$

$\eta$ : subthreshold slope factor  
 $K = W/L$

# Low-Power BGR

- Switching the BGR can be used to reduce the quiescent current.
- Self-cascode cells with transistors in sub-threshold region can reduce the quiescent current too.
- A resistor-less BGR circuit.



$$V_{REF} = \frac{V_{BE,Q1}}{2} + nV_T \ln\left(\frac{S_3 S_5 S_7 S_9}{S_2 S_4 S_6 S_8}\right)$$

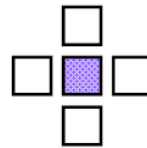
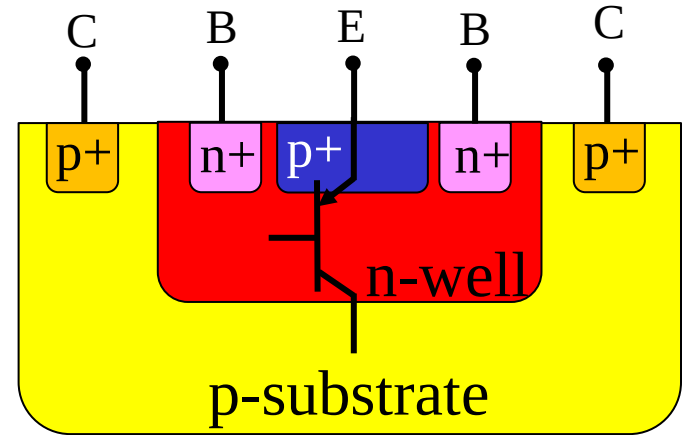
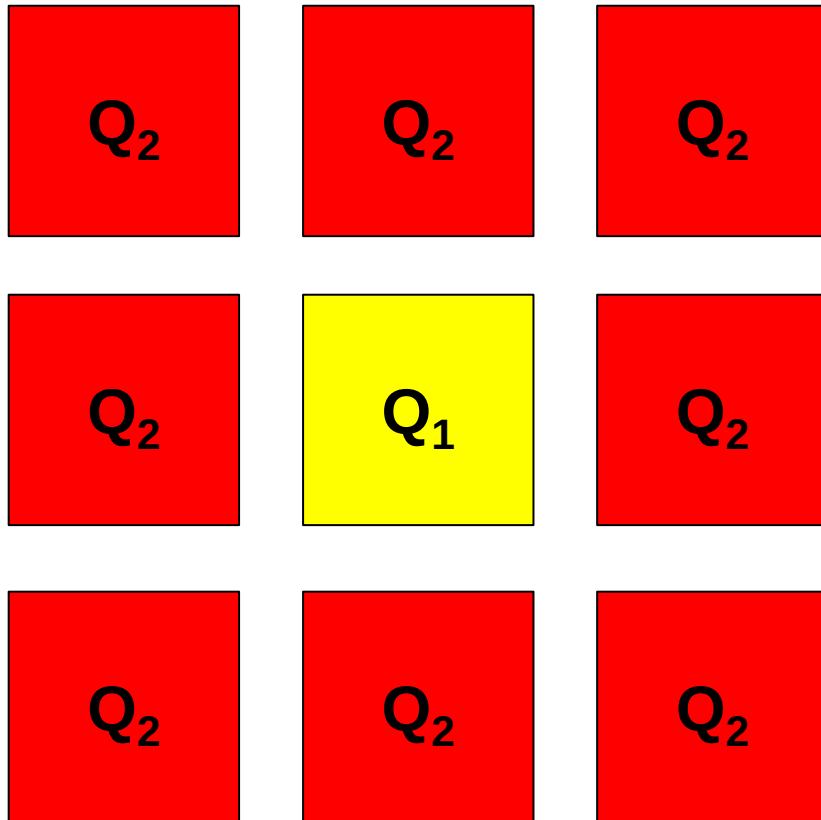
$n$ : subthreshold slope factor

Voltage & Current References



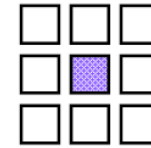
# Layout Issues

- $m$  is often equal to 8 for ease of layout matching.
- A parasitic bipolar is often used in standard CMOS technologies.
- Higher  $m$  is preferred to counteract offset voltage due to mismatch.
- In the industry,  $N=24$  is commonly used.



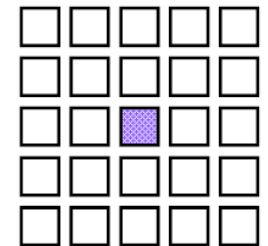
$$Q_1:Q_2 = 4:1$$

$$V_T \ln(4) = 35.9 \text{ mV}$$



$$Q_1:Q_2 = 8:1$$

$$V_T \ln(8) = 53.8 \text{ mV}$$



$$Q_1:Q_2 = 24:1$$

$$V_T \ln(24) = 82.2 \text{ mV}$$

Voltage & Current References

- $$V_{\text{BE}}(T) = V_{\text{BG}} - (V_{\text{BG}} - V_{\text{BE0}}) \frac{T}{T_0} - (\eta - \alpha) V_T \ln \frac{T}{T_0}$$

$$V_{NL} \cong V_{BE, Q_3}(T) - V_{BE, Q_{1,2}}(T) = V_T \ln \frac{T}{T_0}$$

$$V_{\text{ref}} = \frac{R_3}{R_1} \left( \frac{R_1 \ln(N)}{R_0} V_T + V_{BE} + \frac{R_1}{R_{4,5}} V_{NL} \right) \rightarrow R_{4,5} = \frac{R_1}{\eta - 1}$$
[illegible]

Figure 1 is a graph showing Voltage versus Temperature. The Y-axis is labeled "Voltage" and the X-axis is labeled "Temperature". The graph illustrates the components and the resulting bandgap voltage reference (VREF) for the proposed design.

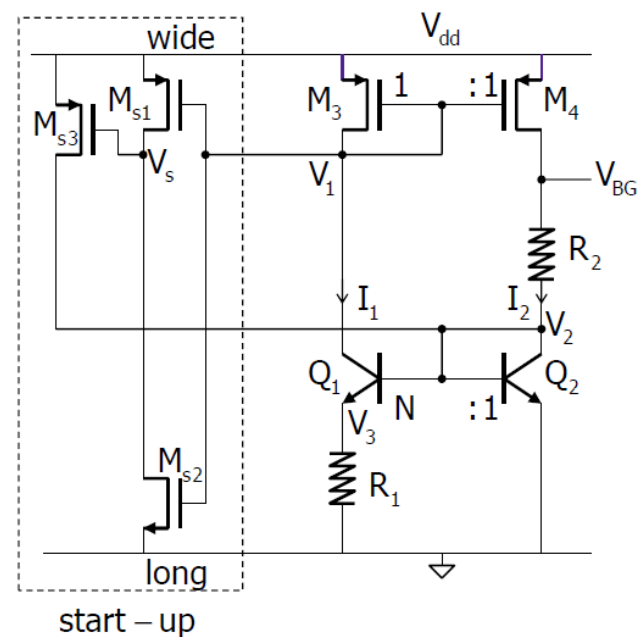
The components shown are:

- (1) CTAT voltage (dashed line, decreasing)
- (2) Scaled PTAT voltage (dash-dot line, increasing)
- (3) Non-linearity (solid line, parabolic)

The conventional bandgap VREF is represented by the dashed line labeled (1) + (2) Conventional Bandgap VREF. The proposed bandgap VREF is represented by the solid line labeled (1) + (2) - (3) Proposed Bandgap VREF, which is the difference between the conventional VREF and the non-linearity component.

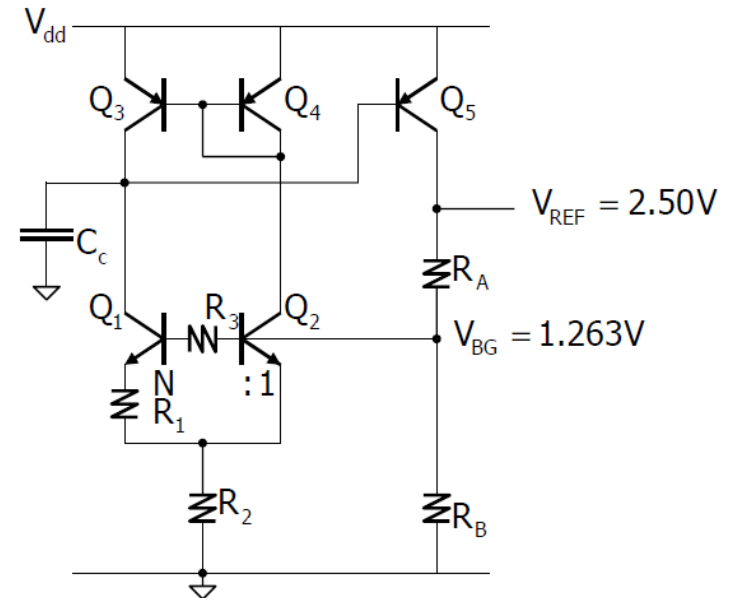
# Startup Circuit

- Circuit has two stable states, so we need a startup.
- When BGR is shut down,  $V_1 = V_{dd}$  and  $V_s = 0$ , turning on  $M_{s3}$  that pumps current to  $Q_2$ .
- $Q_2$  turns on  $Q_1$ , then  $M_3$ , and then  $M_4$ . The BGR works properly.
- When  $M_3$  conducts,  $V_1$  turns on  $M_{s1}$  hard and puts itself into the linear region with a very small  $V_{sd1}$ .
- A small  $V_{sd1}$  shuts off  $M_{s3}$  and isolates  $M_{s3}$  from the BGR.



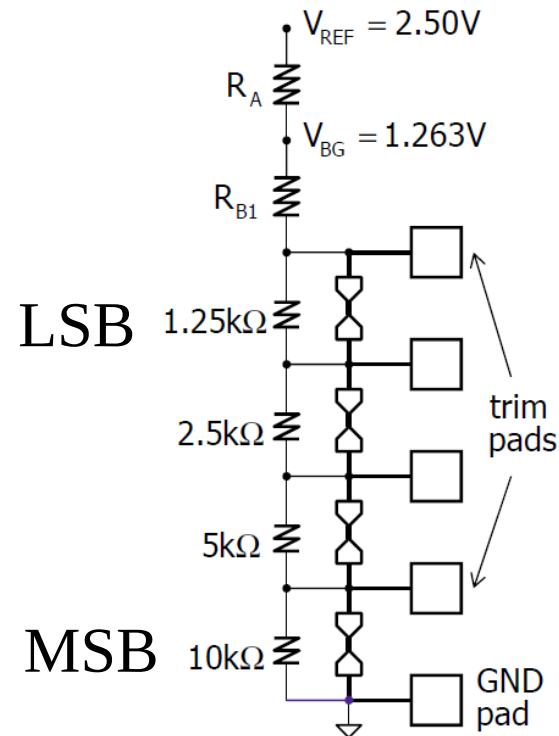
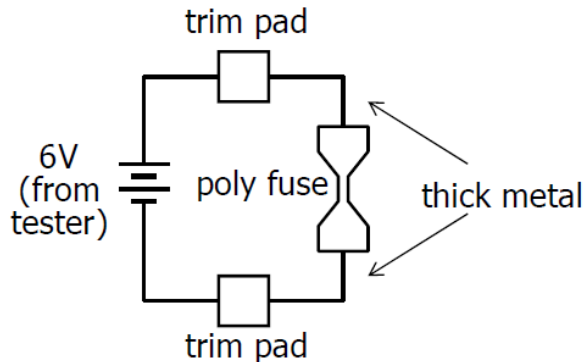
# BGR Trimming

- A very accurate  $V_{REF}$  will need to trim for both zero TC and accurate initial value.
- The zero TC point is determined by  $R_1:R_2$ , while the initial accuracy of  $V_{REF}$  (at room temp.) is determined by  $R_A:R_B$ .
- Trimming individual BGR for zero TC is too time consuming.
- In practice, the zero TC point is determined by fabricating and measuring the BGRs a few times using the same process.
- After finding the “optimal” ratio of  $R_1:R_2$ , this is then fixed for mass production and individual BGR will only be trimmed for initial accuracy.



# Trimming Circuit

- Trimming is usually done during wafer testing. A poly fuse of around  $100\Omega$  needs 60mA to be blown open.
- $R_{\text{LSB}} \gg R_{\text{poly}}$ : blow fuse to trim voltage up.



# BGR Specifications: Accuracy

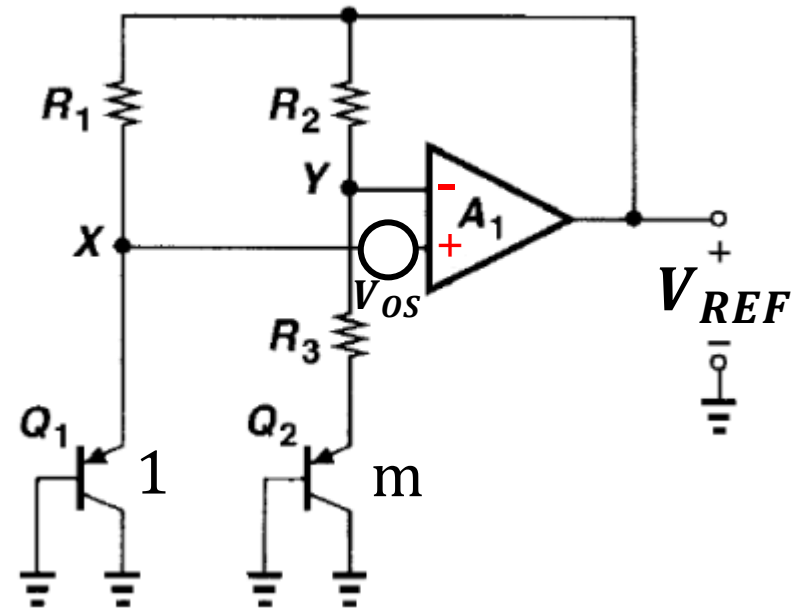
- A bandgap is said to have 10% accuracy if  $V_{REF}$  is within  $\pm 10\%$  of the desired value.
- Inaccuracies result from
  - Finite gain of opamps
  - Systematic mismatches
  - Random mismatches of resistors, MOS transistors and BJTs
  - Opamp offset

$$V_X = V_Y + V_{OS}$$

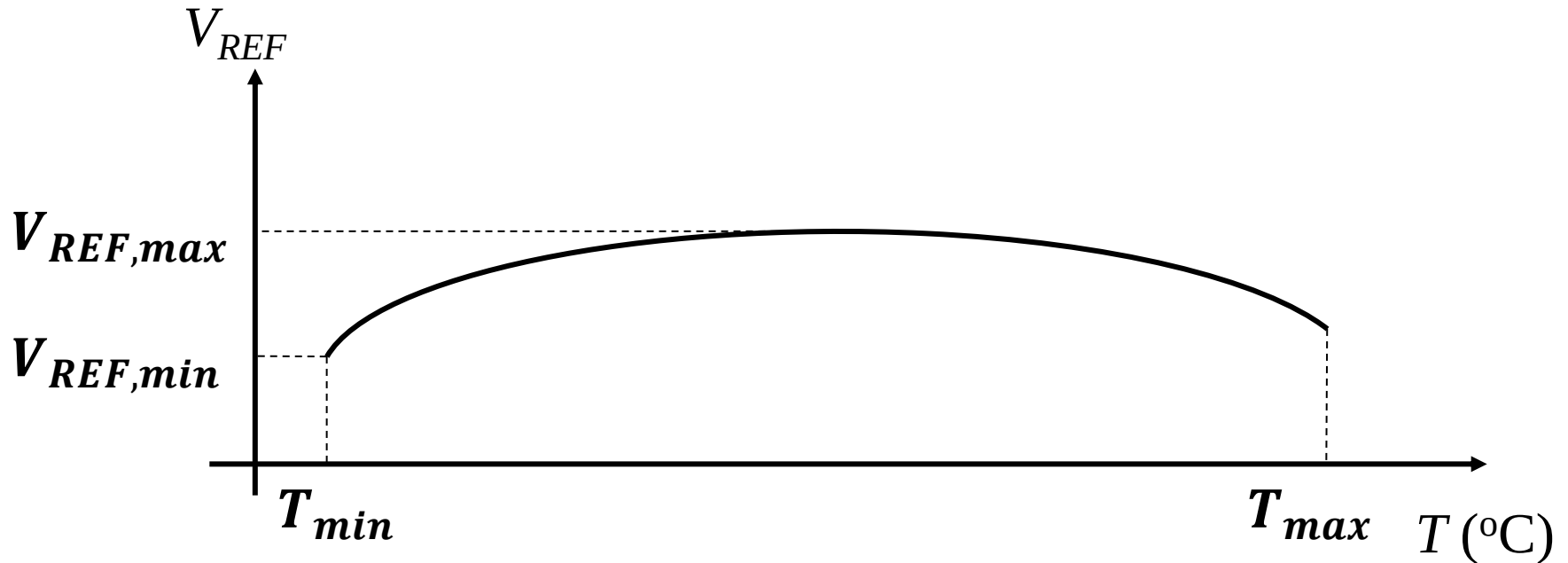
$$\Delta V_{EB} = V_{EB1} - V_{EB2} = IR_3 + V_{OS}$$

$$I = \frac{V_T}{R_3} \ln(m) - \frac{V_{OS}}{R_3}$$

$$V_{REF} = V_{EB1} + V_T \ln(m) \frac{R_1}{R_3} - V_{OS} \frac{R_1}{R_3}$$



# BGR Specifications: Temp. Coefficient



- $$TC = \frac{V_{REF,max} - V_{REF,min}}{\left(\frac{V_{REF,max} + V_{REF,min}}{2}\right) T_{max} - T_{min}} \frac{1}{T_{max} - T_{min}}$$
- TC is specified in ppm/ $^{\circ}\text{C}$



# BGR Specifications: Others

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- **Line Regulation**

- A measure of the circuit's ability to maintain the specified output voltage with varying input voltage
- $\text{Line Regulation} = \frac{\Delta V_{REF}}{\Delta V_{DD}}$

- **Load Regulation**

- A measure of the circuit's ability to maintain the specified output voltage under varying load conditions
- $\text{Load Regulation} = \frac{\Delta V_{REF}}{\Delta I_O}$

- **Power Supply Rejection**

- A measure of the circuit's ability to prevent the output voltage fluctuating caused by input voltage variations
- $\text{PSR} = \frac{V_{o,ripple}}{V_{i,ripple}}$

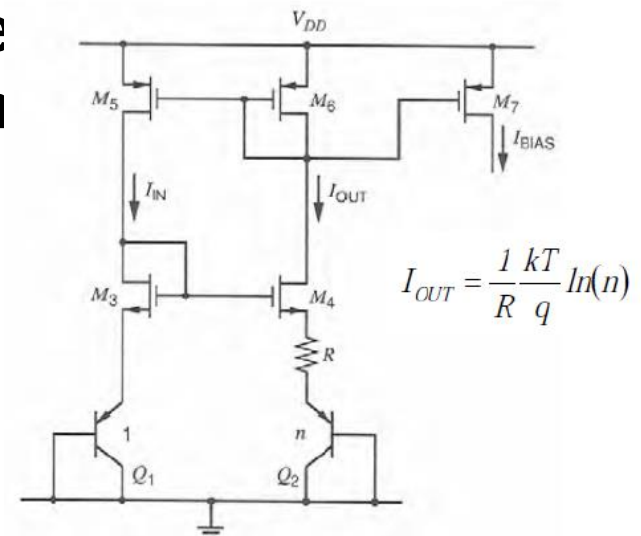
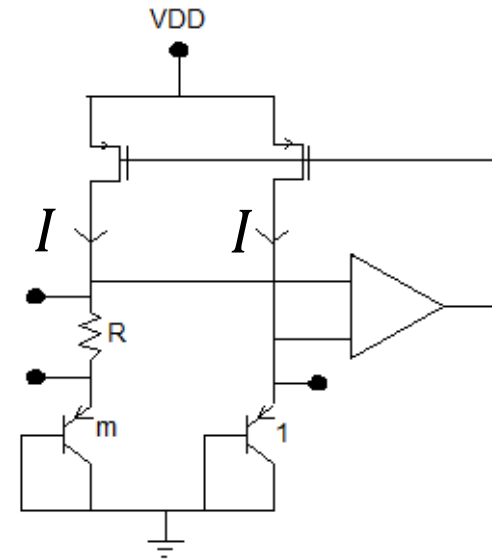
Voltage & Current References



# Current References: PTAT Current

$$I = \frac{\Delta V_{EB}}{R} = \frac{V_T}{R} \ln(m)$$

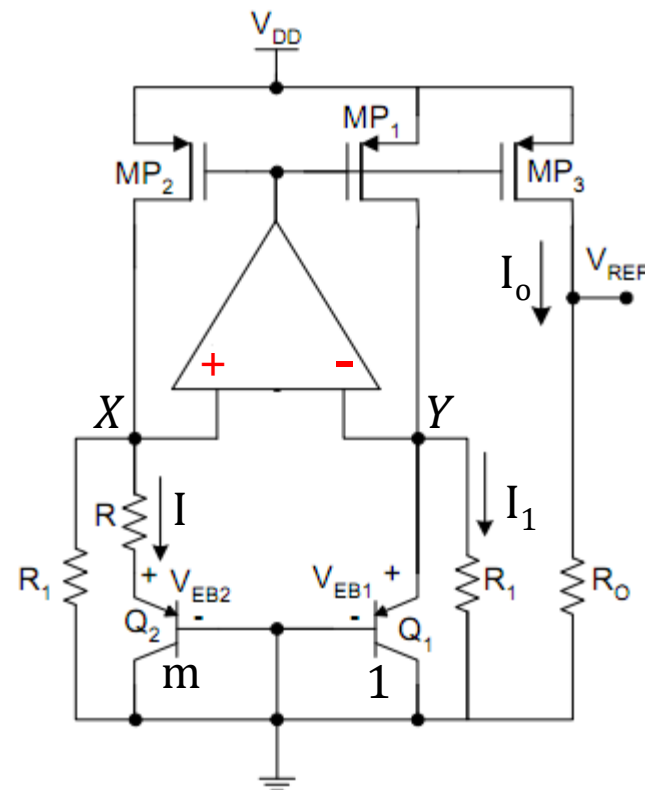
- Easy to generate when a BG is available.
- Inversely proportional to resistor value. (pseudo-PTAT)
- Current is supply independent.
- PTAT current can compensate for mobility degradation at high temperatures.
- Can be done with no opamp.



# Current References: Poly Current

$$I_o = \frac{V_{REF}}{R_o} = \frac{V_{EB1}}{R_1} + \frac{V_T}{R} \ln(m)$$

- Easy to generate when PTAT and CTAT currents are added in the BG.
- Inversely proportional to resistor value. (pseudo-ZTAT)
- Current is supply independent.
- Poly current can be used to generate reference voltages and constant swings.

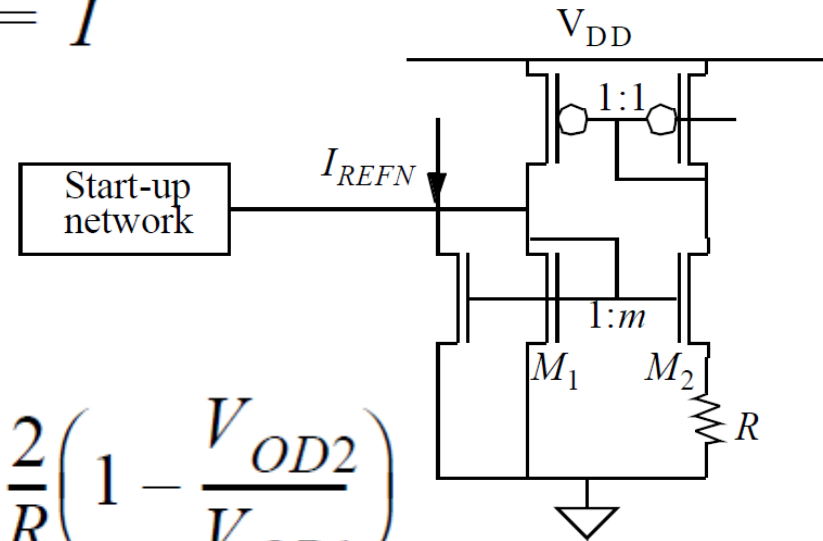


# Current References: Constant-gm Current

$$I_2 = \frac{\Delta V_{GS}}{R} = \frac{\Delta V_{OD}}{R} = I_1 = I$$

$$g_m = \frac{2I}{V_{OD}}$$

$$g_{m1} = \frac{2I}{V_{OD1}} = \frac{2\Delta V_{OD}}{RV_{OD1}} = \frac{2}{R} \left( 1 - \frac{V_{OD2}}{V_{OD1}} \right)$$



$$\frac{V_{OD2}}{V_{OD1}} = \sqrt{\frac{W_1}{W_2}} = \frac{1}{\sqrt{m}}$$

$$g_{m1} = \frac{2}{R} \left( 1 - \frac{1}{\sqrt{m}} \right)$$

# Current References: Constant Current

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- **Can be obtained by:**
  - Using an external resistor.
  - Using trimming.
  - Using resistors with opposite temperature coefficients.

